

Biotechnology and Food Engineering

Process Biotechnology

Spring 2026

Tutorial #5 - Continuous fermentation – Solutions

Question 1

A factory aims to produce a certain toxin secreted by yeast in a 1 m³ fermenter. The fermentation starts as batch growth. The initial glucose concentration at the beginning of the batch fermentation is 4% and the initial biomass concentration is 2 $\frac{gr}{lit}$. The growth yield is 0.46 $\frac{gX}{gS}$, the production yield is 210.5 mg toxin to gr biomass, the maximal growth rate is 0.2 per hour and Monod constant (K_s) is 1000 ppm.

After 12.5 hours of fermentation, when the biomass concentration in the system reach to 1.9%, the fermentation operator switch to continuous fermentation by feeding the fermenter with fresh medium in a certain flow rate, F, and pumping out part of the fermenter content in the same flow rate F.

- A. What is the growth rate of the yeast in the exact moment when the batch fermentation is changed to continuous?**
- B. Assuming that during the steady state of the continuous fermentation we maintain the same growth rate and biomass concentration as was at the end of the batch fermentation:**
- 1) If we supply molasses with solid concentration of 44% that contain 67% glucose, what should be the molasses concentration in the fresh medium enters the continuous fermentation?**
 - 2) The product flow rate leaving the continuous fermentation is limited by the maximal capacity of the separation system, which is 500 $\frac{grP}{hr}$. According to this given data, and in order to meet the requirement stated at the beginning of this section (to keep the same growth rate and biomass concentration as was at the point of changing between batch and continuous fermentation),**

calculate the parameters that should be fixed so that the product flow rate would fit to the separation system capacity.

Solution

Given:

<u>Batch</u>	<u>Continuous</u>
$V = 1 \text{ m}^3 = 1000 \text{ liter}$	$X_0 = 1.9\% = 19 \frac{gX}{lit}$
$S_0 = 4\% = 40 \frac{gS}{lit}$	$molasses = 44\% \text{ solid}$
$X_0 = 2 \frac{gX}{lit}$	$molasses_{solid} = 67\% \text{ glucose}$
$Y_{\frac{X}{S}} = 0.46 \frac{gX}{gS}$	
$Y_{\frac{P}{X}} = 210.5 \frac{mgP}{gX} = 0.2105 \frac{gP}{gX}$	$G'_{p(max)} = 500 \frac{gP}{hr}$
$\mu_{max} = 0.2 \frac{1}{hr}$	
$K_s = 1000 \text{ ppm} = 1 \frac{g}{lit}$	
$t = 12.5 \text{ hr}$	

A. $\mu_{t=12.5 \text{ hr}} = ?$

There are 2 options to find the growth rate:

- I. Growth equation: $\ln \frac{x_t}{x_0} = \mu \cdot t$
- II. Monod equation: $\mu = \mu_{max} \cdot \left(\frac{S}{K_s + S} \right)$

But we can't be sure that $t = t_{log}$ so we can't use the growth equation. therefore, we will use Monod equation.

We first need to find S_t at 12.5 hr. We will use $Y_{\frac{X}{S}}$:

$$Y_{\frac{X}{S}} = \frac{X_t - X_0}{S_0 - S_t} = \frac{(19 - 2) \frac{gX}{lit}}{(40 - S_t) \frac{gS}{lit}} = 0.46 \frac{gX}{gS}$$

$$S_t = 3.04 \frac{gS}{lit}$$

Now we can place numbers in Monod equation and find μ :

$$\mu = 0.2 \frac{1}{hr} \cdot \left(\frac{3.04 \frac{gS}{lit}}{1 \frac{g}{lit} + 3.04 \frac{gS}{lit}} \right) = 0.15 \frac{1}{hr}$$

$$\mu = 0.15 \frac{1}{hr}$$

B.

(1) $m_0 = ?$

In order to find the molasses concentration in the feed we need to find what is S_0 .

The fermentation is standard so we can use the biomass balance:

$$\frac{dx}{dt} = \left(\frac{F}{V} \right) \cdot X_0 + \mu \cdot X_F - \alpha \cdot X_F - \left(\frac{F}{V} \right) \cdot X_F$$

- The feed is sterile so $X_0 = 0$.
- The death rate is negligible so $\alpha \ll \mu$.
- During steady state $\frac{dx}{dt} = 0$.

$$\text{Therefore, } \mu = \frac{F}{V} = D$$

From the substrate balance:

$$\frac{ds}{dt} = \left(\frac{F}{V} \right) \cdot S_0 - \frac{\mu \cdot X_F}{Y_{x/s}} - \frac{q_p \cdot X_F}{Y_{p/s}} - m_s \cdot X_F - \left(\frac{F}{V} \right) \cdot S_F$$

- We previously found that $\mu = \frac{F}{V} = D$.
- The sizes of $\frac{q_p \cdot X_F}{Y_{p/s}}$ and $m_s \cdot X_F$ are negligible.
- During steady state $\frac{ds}{dt} = 0$.

$$\text{Therefore, } \mu \cdot S_0 - \frac{\mu \cdot X_F}{Y_{x/s}} - \mu \cdot S_F = 0 \text{ and we get the final equation: } X_F = Y_{x/s} \cdot (S_0 - S_F).$$

Now we can place numbers to find S_0 :

$$19 \frac{gX}{lit} = 0.46 \frac{gX}{g_{glucose}} \cdot \left(S_0 - 3.04 \frac{g_{glucose}}{lit} \right)$$

$$S_0 = 44.3 \frac{g_{glucose}}{lit}$$

Now we need to go back to the molasses parameters:

$$m_0 = \frac{44.3 \frac{g_{glucose}}{lit}}{0.67 \frac{g_{solid}}{g_{molasses}} \cdot 0.44 \frac{g_{glucose}}{g_{solid}}}$$

$$m_0 = 150.3 \frac{g_{molasses}}{lit}$$

(2) We need to find what is the production flow rate (G_p) and if it is too high than what are the options to reduce it.

The maximal production flow rate G_p is dictated by the capacity of the separation system.

$$G'_{p(max)} = F \cdot P_F = 500 \frac{gP}{hr}$$

We first need to find the production flow rate (G_p) of the fermentation, before considering the separation process constrain. To do so, we need to calculate F and P_F .

The product concentration (P_F) can be found by using the product balance:

$$\frac{dp}{dt} = \left(\frac{F}{V} \right) \cdot P_0 + q_p \cdot X_F - K \cdot P_F - \left(\frac{F}{V} \right) \cdot P_F$$

- No product in the feed so $P_0 = 0$.
- K is negligible.
- During steady state $\frac{dp}{dt} = 0$.
- $q_p = \mu \cdot Y_{p/x}$ and $\mu = \frac{F}{V} = D$

Therefore, we get the following equation: $P_F = Y_{p/x} \cdot X_F$, from which we can calculate P_F :

$$P_F = 0.2105 \frac{gP}{gX} \cdot 19 \frac{gX}{lit} = 4 \frac{gP}{lit}$$

To find the flow rate (F) we can use the correlation between the specific growth rate and the dilution rate:

$$\mu = D = \frac{F}{V}$$

$$F = \mu \cdot V = 0.15 \text{ hr}^{-1} \cdot 1000 \text{ lit}$$

$$F = 150 \frac{\text{lit}}{\text{hr}}$$

Now we can find G_p :

$$G_p = 150 \frac{\text{lit}}{\text{hr}} \cdot 4 \frac{\text{gP}}{\text{lit}} = 600 \frac{\text{gP}}{\text{hr}}$$

We can see that the production flow rate is too high. We can reduce it in two ways: by reducing either F or P_F . It is not possible to change P_F since X_F and $Y_{P/X}$ are as required in the fermentation, so the only real option is to reduce F .

If we reduce F it means that we also need to change V in order to maintain the growth rate.

What are the new F and V ?

$$F = \frac{G'_{p(max)}}{P_F} = \frac{500 \frac{\text{gP}}{\text{hr}}}{4 \frac{\text{gP}}{\text{lit}}} = 125 \frac{\text{lit}}{\text{hr}}$$

$$V = \frac{F}{\mu} = \frac{125 \frac{\text{lit}}{\text{hr}}}{0.15 \frac{1}{\text{hr}}} = 833.3 \text{ lit}$$

In summary, in order to ensure a proper flow rate that meet the separation process constrain, we need to change the flow rate to **$F = 125 \frac{\text{lit}}{\text{hr}}$** and the fermenter volume to **$V = 833.3 \text{ lit}$** .

Question 2

A factory performs continuous growth of yeast in a 1250 liter fermenter. The glucose concentration in the fresh feed flow is 4% and the growth yield is $0.48 \frac{gX}{gS}$. In a standard fermentation and in steady state, the substrate concentration in the fermenter is $8.75 \frac{gS}{lit}$. The maximal growth rate is 0.218 per hour and Monod constant (K_s) is 750 ppm.

When the factory engineer returned after a long weekend vacation, he surprisingly found that the flow rate controller, which its role is to control the flow rate of the fresh feed flow (F) into the fermenter, is broken, and that the flow rate was significantly greater than required. As a results, the fermenter enters to a rinsing process and now the biomass concentration in the leaving flow is only 100 ppm. From the controller parameters it is known that the change in the feed flow rate occurred exactly 42.5 hours before the engineer noticed the malfunction.

- A. In what percentages was the flow rate deviated when the error occurred, relatively to the standard flow rate in that process?**
- B. What are the steps that the engineer should perform in order to bring the fermentation back to normal, as soon as possible?**

Solution:

Given:

<u>Normal</u>	<u>Malfunctioning</u>
$V = 1250 \text{ lit}$	$X_{F'} = 1000 \text{ ppm} = 0.1 \frac{gX}{lit}$
$S_0 = 4\% = 40 \frac{gS}{lit}$	$t = 42.5 \text{ hr}$
$Y_{\frac{x}{s}} = 0.48 \frac{gX}{gS}$	
$S_F = 8.75 \frac{gS}{lit}$	
$\mu_{max} = 0.218 \frac{1}{hr}$	

$$K_s = 750 \text{ ppm} = 0.75 \frac{\text{g}}{\text{lit}}$$

A. $\frac{F'}{F} = ?$

We need to find the flow rate during normal fermentation (F) and after the malfunction occurred (F').

First, we will find F , using the biomass balance:

$$\frac{dx}{dt} = \left(\frac{F}{V}\right) \cdot X_0 + \mu \cdot X_F - \alpha \cdot X_F - \left(\frac{F}{V}\right) \cdot X_F$$

Assumptions:

- Steady state $\frac{dx}{dt} = 0$
- Sterile feed $X_0 = 0$
- $\alpha \ll \mu$

So the final equation is: $\mu = \frac{F}{V}$

In order to find μ we will use the Monod equation:

$$\mu = \mu_{max} \cdot \left(\frac{S}{K_s + S}\right) = 0.218 \frac{1}{hr} \cdot \left(\frac{8.75 \frac{\text{gS}}{\text{lit}}}{0.75 \frac{\text{g}}{\text{lit}} + 8.75 \frac{\text{gS}}{\text{lit}}}\right) = 0.2 \frac{1}{hr}$$

The fermenter volume is given so:

$$F = \mu \cdot V = 0.2 \frac{1}{hr} \cdot 1250 \text{ lit} = 250 \frac{\text{lit}}{hr}$$

Second, we will find F' , using the biomass balance:

$$\frac{dx}{dt} = \left(\frac{F}{V}\right) \cdot X_0 + \mu \cdot X_F - \alpha \cdot X_F - \left(\frac{F}{V}\right) \cdot X_F$$

Assumptions:

- Sterile feed $X_0 = 0$
- $\alpha \ll \mu$

Note! The fermentation is not in steady state, so we get the following:

$$\frac{dx}{dt} = \mu \cdot x - \left(\frac{F}{V}\right) \cdot x = x \cdot (\mu - D)$$

We change sides to obtain:

$$\int_{x_1}^{x_2} \frac{dx}{x} = \int_{t_1}^{t_2} (\mu - D) dt$$

$$\ln \frac{x_2}{x_1} = (\mu_{max} - D) \Delta t$$

* When a fermenter is in rinsing state then $\mu = \mu_{max}$.

What are x_1 and x_2 ?

x_2 is the biomass concentration at $t = 42.5 \text{ hr}$ (when the error was found by the engineer).

x_1 is the biomass concentration just before the rinsing process begun. This is “time zero” so eventually x_1 is x_F which can be found using the substrate balance:

$$\frac{ds}{dt} = \left(\frac{F}{V}\right) \cdot S_0 - \frac{\mu \cdot X_F}{Y_{\frac{x}{s}}} - \frac{q_p \cdot X_F}{Y_{\frac{p}{s}}} - m_s \cdot X_F - \left(\frac{F}{V}\right) \cdot S_F$$

Assumptions:

- Steady state $\frac{ds}{dt} = 0$
- Negligible $\frac{q_p \cdot X_F}{Y_{\frac{p}{s}}}$ (biomass production)

➤ $m_s \cdot X_F$ is negligible during logarithmic growth

So we can work with the final equation:

$$X_F = Y_{\frac{x}{s}} \cdot (S_0 - S_F) = 0.48 \frac{gX}{gS} \cdot (40 - 8.75) \frac{gS}{lit}$$

$$X_F = 15 \frac{gX}{lit}$$

Now we can go back to the equation that was developed and place numbers:

$$\ln \frac{x_2}{x_1} = (\mu_{max} - D) \Delta t$$

$$\ln \left(\frac{0.1 \frac{gX}{lit}}{15 \frac{gX}{lit}} \right) = \left(0.218 \frac{1}{hr} - D \right) \cdot 42.5 \text{ hr}$$

$$D = 0.335 \frac{1}{hr}$$

$$D = \frac{F'}{V}$$

$$F' = 0.335 \frac{1}{hr} \cdot 1250 \text{ lit}$$

$$F' = 412.5 \frac{lit}{hr}$$

Now that we have both F and F' we can calculate the ratio:

$$\frac{F'}{F} = \frac{412.5 \frac{lit}{hr}}{250 \frac{lit}{hr}} = 1.65$$

The flow rate increased in 65%.

B. The steps that the engineer should perform are:

- 1) *Return the flow rate to its normal rate F .*
- 2) *To increase X_F the engineer should add more substrate, according to $X_F = Y_{\frac{x}{s}} \cdot (S_0 - S_F)$.*
- 3) *Return to batch fermentation until the yeast grow enough so the fermentation can be changed back to continuous.*

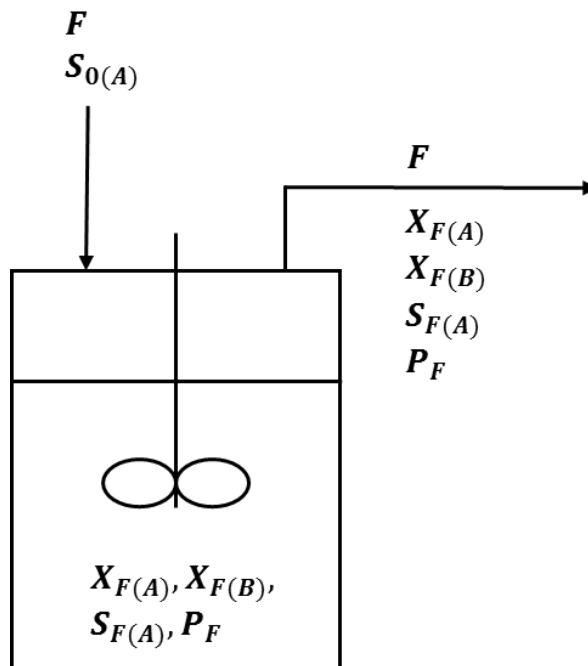
Question 3

A research laboratory performs continuous fermentation of two different microorganisms, yeast and bacteria, growing simultaneously in a 500 liter fermenter. The fermenter is supplied with a fresh feed in flow rate of $200 \frac{\text{lit}}{\text{hr}}$, that contains a certain pentose (marked as $S_{0(A)}$) in concentration of 2.6%.

Only the yeast can utilize the pentose and the growth yield on it is $0.54 \frac{gX}{gS}$. The maximal growth rate of the yeast on this pentose is 0.5 hr^{-1} and the given Monod constant is 250 ppm. The yeast produce alcohol in yield of $2 \frac{gP}{gX}$, while **this alcohol serves as the substrate for the bacteria growth in this fermenter.**

The bacteria grow on the alcohol produced by the yeast, in yield of $0.7 \frac{gX}{gS}$. The maximal growth rate of the bacteria on this alcohol is 0.65 hr^{-1} and Monod constant is $0.8 \frac{gr}{lit}$.

The flow exiting the fermenter contains: yeast ($X_{F(A)}$), bacteria ($X_{F(B)}$), pentose leftovers that were not used by the yeast ($S_{F(A)}$) and the alcohol leftovers that were not used by the bacteria (P_F). For your convenience see the process scheme below:



What is the bacteria concentration in the exiting flow ($X_{F(B)}$), during steady state?

Solution:

Given:

<u>Fermentation</u>	<u>Yeast (A)</u>	<u>Bacteria (B)</u>
$V = 500 \text{ lit}$	$Y_{\frac{x}{s}} = 0.54 \frac{gX}{gS}$	$Y_{\frac{x}{s}} = 0.7 \frac{gX}{gS}$
$F = 200 \frac{\text{lit}}{\text{hr}}$	$\mu_{max} = 0.5 \frac{1}{\text{hr}}$	$\mu_{max} = 0.65 \frac{1}{\text{hr}}$
$S_{0(A)} = 2.6\% = 26 \frac{gS}{\text{lit}}$	$K_s = 250 \text{ ppm} = 0.25 \frac{g}{\text{lit}}$	$K_s = 0.8 \frac{g}{\text{lit}}$
	$Y_{\frac{p}{x}} = 2 \frac{gP}{gX}$	$S_B = P_A = \text{alcohol}$

$X_{F(B)} = ?$

Note - the solution of this question is not standard since the fermentation is continuous thus the balances will be treated different.

There are 2 ways of solution.

Option I:

First, we will use the bacteria biomass balance, where $X_{F(B)}$ appears:

$$\frac{dx}{dt} = \left(\frac{F}{V}\right) \cdot X_{0(B)} + \mu_B \cdot X_{F(B)} - \alpha \cdot X_{F(B)} - \left(\frac{F}{V}\right) \cdot X_{F(B)}$$

Assumptions:

- Steady state $\frac{dx}{dt} = 0$
- Sterile feed $X_{0(B)} = 0$
- $\alpha \ll \mu$

So the final equation is: $\mu_B = \frac{F}{V} = D$.

We can do the same for the yeast biomass balance to get a similar equation: $\mu_A = \frac{F}{V} = D$.

Now we will perform balance on the alcohol, which is the yeast product and the bacteria substrate, so we need to take both into consideration:

$$\frac{ds_B}{dt} = \left(\frac{F}{V}\right) \cdot S_{0(B)} - \frac{\mu_B \cdot X_{F(B)}}{Y_{\frac{x}{s}(B)}} - \frac{q_p \cdot X_F}{Y_{\frac{p}{s}}} - m \cdot X_{F(B)} - \left(\frac{F}{V}\right) \cdot S_{F(B)} + [q_{p(A)} \cdot X_{F(A)} - K \cdot S_B]$$

Explanation of the different compartments of the equation above:

$\frac{\mu_B \cdot X_{F(B)}}{Y_{\frac{x}{s}(B)}}$ represents the alcohol consumed by the bacteria for its growth.

$\frac{q_p \cdot X_F}{Y_{\frac{p}{s}}}$ represents the alcohol consumed by the bacteria for product production.

$m \cdot X_{F(B)}$ represents the alcohol consumed for maintenance.

$q_{p(A)} \cdot X_{F(A)}$ is the alcohol production rate by the yeast.

$K \cdot S_B$ is the alcohol decomposition due to chemical reactions.

Assumptions that can be made:

- Steady state $\frac{ds}{dt} = 0$
- $S_{0(B)} = 0$
- $\frac{q_p \cdot X_F}{Y_{p_s}}$, $m_s \cdot X_F$ and $K \cdot S_B$ are negligible

So we get: $q_{p(A)} \cdot X_{F(A)} - \frac{\mu_B \cdot X_{F(B)}}{Y_{x_s(B)}} - \left(\frac{F}{V}\right) \cdot S_{F(B)} = 0$

We know that: $q_{p(A)} = \mu_A \cdot Y_{\frac{p}{x_A}}$, and that $\mu_A = \mu_B = D$

Finally, we get:

$$X_{F(B)} = Y_{\frac{x}{s}(B)} \cdot (Y_{\frac{p}{x_A}} \cdot X_{F(A)} - S_{F(B)})$$

Before we can use this equation, we need to find $S_{F(B)}$ and $X_{F(A)}$.

$S_{F(B)}$ can be found from the bacteria Monod equation:

$$\mu = \mu_{max} \cdot \left(\frac{S}{K_s + S}\right) \rightarrow S_{F(B)} = \left(\frac{\mu \cdot K_{s(B)}}{\mu_{max(B)} - \mu}\right)$$

$$\mu = D = \frac{F}{V} = \frac{200 \frac{lit}{hr}}{500 lit} = 0.4 \frac{1}{hr}$$

$$S_{F(B)} = \left(\frac{0.4 \frac{1}{hr} \cdot 0.8 \frac{g}{lit}}{(0.65 - 0.4) \frac{1}{hr}}\right)$$

$$S_{F(B)} = 1.28 \frac{g}{lit}$$

Now, to find $X_{F(A)}$ we will perform balance on the yeast substrate, the pentose:

$$\frac{ds_A}{dt} = \left(\frac{F}{V}\right) \cdot S_{0(A)} - \frac{\mu_A \cdot X_{F(A)}}{Y_{x_s(A)}} - \frac{q_p \cdot X_{F(A)}}{Y_{p_s(A)}} - m \cdot X_{F(A)} - \left(\frac{F}{V}\right) \cdot S_{F(A)}$$

Assumptions:

- Steady state $\frac{ds}{dt} = 0$
- $\frac{q_p \cdot X_F}{Y_p^s}$ and $m_s \cdot X_F$ are negligible.
- $\mu = \left(\frac{F}{V}\right)$

So we get: $X_{F(A)} = Y_{\frac{x}{s}(A)} \cdot (S_{0(A)} - S_{F(A)})$

We have all the parameters except $S_{F(A)}$ that can be found from Monod equation for the yeast (similar to what we did above for the bacteria):

$$S_{F(A)} = \left(\frac{0.4 \frac{1}{hr} \cdot 0.25 \frac{g}{lit}}{(0.5 - 0.4) \frac{1}{hr}} \right)$$

$$S_{F(A)} = 1 \frac{g}{lit}$$

Place numbers in the equation we developed:

$$X_{F(A)} = Y_{\frac{x}{s}(A)} \cdot (S_{0(A)} - S_{F(A)}) = 0.54 \frac{gX}{gS} \cdot (26 - 1) \frac{gS}{lit}$$

$$X_{F(A)} = 13.5 \frac{gX}{lit}$$

Finally, we can place $S_{F(B)}$ and $X_{F(A)}$ in the equation to find $X_{F(B)}$:

$$X_{F(B)} = Y_{\frac{x}{s}(B)} \cdot (Y_{\frac{p}{x_A}} \cdot X_{F(A)} - S_{F(B)})$$

$$X_{F(B)} = 0.7 \frac{gX}{g} \cdot \left(2 \frac{g}{gX} \cdot 13.5 \frac{gX}{lit} - 1.28 \frac{g}{lit} \right)$$

*Note – $S=P$ in this case so the unit were written generally as ‘g’.

$$X_{F(B)} = 18 \frac{gX}{lit}$$

Option II:

We can perform balance on the bacteria substrate (the alcohol):

$$\frac{ds_B}{dt} = \left(\frac{F}{V}\right) \cdot S_{0(B)} - \frac{\mu_B \cdot X_{F(B)}}{Y_{\frac{x}{s}(B)}} - \frac{q_p \cdot X_F}{Y_{\frac{p}{s}}} - m \cdot X_{F(B)} - \left(\frac{F}{V}\right) \cdot S_{F(B)}$$

Assumptions:

- Steady state $\frac{ds}{dt} = 0$
- $\frac{q_p \cdot X_F}{Y_{\frac{p}{s}}}$, $m \cdot X_F$ and $K \cdot S_B$ are negligible

The bacteria substrate is not coming from the feed but from the alcohol production by the yeast. We can say that $\left(\frac{F}{V}\right) \cdot S_{0(B)} = \left(\frac{F}{V}\right) \cdot P_{F(A)}$.

So we get: $\left(\frac{F}{V}\right) \cdot P_{F(A)} - \frac{\mu_B \cdot X_{F(B)}}{Y_{\frac{x}{s}(B)}} - \left(\frac{F}{V}\right) \cdot S_{F(B)} = 0$

From the previous biomass balances for the yeast and bacteria we know that $\mu = \frac{F}{V} = D$.

Finally, we get the following equation:

$$X_{F(B)} = Y_{\frac{x}{s}(B)} \cdot (P_{F(A)} - S_{F(B)})$$

$Y_{\frac{x}{s}(B)}$ is given in the question ($0.7 \frac{gX}{gS}$).

$S_{F(B)}$ can be found from Monod equation on the bacteria as was done in way I ($1.28 \frac{g}{lit}$).

$P_{F(A)}$ can be found by using the product balance for the yeast:

$$\frac{dp}{dt} = \left(\frac{F}{V}\right) \cdot P_0 + q_{p(A)} \cdot X_{F(A)} - K \cdot P_{F(A)} - \left(\frac{F}{V}\right) \cdot P_{F(A)}$$

Assumption:

- $\frac{dp}{dt} = 0$
- $P_0 = 0$
- $K \cdot P_{F(A)}$ is negligible

We get that:

$$P_{F(A)} = Y_{\frac{p}{x}(A)} \cdot X_{F(A)}$$

$Y_{\frac{p}{x}(A)}$ is given in the question ($2 \frac{gP}{gX}$)

$X_{F(A)}$ can be found as was done in way I ($13.5 \frac{gX}{lit}$)

We can place these numbers to find $P_{F(A)}$:

$$P_{F(A)} = 2 \frac{gP}{gX} \cdot 13.5 \frac{gX}{lit} = 27 \frac{gP}{lit}$$

Now we can go back to the equation we initially developed, $X_{F(B)} = Y_{\frac{x}{s}(B)} \cdot (P_{F(A)} - S_{F(B)})$

and place numbers:

$$X_{F(B)} = 0.7 \frac{gX}{g} \cdot \left(27 \frac{g}{lit} - 1.28 \frac{g}{lit} \right)$$

$$X_{F(B)} = 18 \frac{gX}{lit}$$